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1. Introduction to Algebraic Structures and Syllabus

An **algebraic structure** is a set together with one or more operations on it that obey fixed rules. The simplest such structure is a **group** — a set with one operation that is associative, has an identity and gives every element an inverse. Groups capture the idea of "symmetry" and "reversible operations", and they appear everywhere: clock arithmetic, the integers, invertible matrices, and even error-correcting codes.

The second half of the unit applies groups to **coding theory**: how to add redundancy to binary information so that errors introduced during transmission can be *detected* and even *corrected*.

1.1 Syllabus of Unit V (Algebraic Structures and Applications)

- Binary operations — **Section 2**
- Semi-groups and groups — **Sections 3-5**
- Subgroups — **Section 6**
- Cosets; Lagrange's theorem — **Sections 8-9**
- Product and quotient semi-groups and groups; normal subgroup — **Section 10**
- Homomorphism — **Section 11**
- Coding of binary information and error detection; group codes; decoding and error correction — **Sections 12-14**
- Solved Question Bank — **Section 15**

2. Binary Operations

Definition (Binary Operation). A **binary operation** on a set S is a function $f : S \times S \rightarrow S$. We usually write the result of the operation on (a,b) as $a * b$. The key point is that the result must again lie in S — this is called **closure**.

Examples.

- Addition $+$ and multiplication $\times$ are binary operations on $\mathbb{Z}$ (and on $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$).
- Subtraction is a binary operation on $\mathbb{Z}$ but **not** on $\mathbb{N}$ (e.g. $3 - 5 = -2 \notin \mathbb{N}$).
- Bitwise XOR $\oplus$ on $B = \{0,1\}$: $0 \oplus 0 = 0$, $0 \oplus 1 = 1$, $1 \oplus 0 = 1$, $1 \oplus 1 = 0$.
- Addition modulo n , written $+_n$, on $\mathbb{Z}_n = \{0,1,\dots,n-1\}$.

Properties an operation may have.

- **Associative:** $(a*b)*c = a*(b*c)$ for all a,b,c .
- **Commutative:** $a*b = b*a$ for all a,b .
- Has an **identity** e : $a*e = e*a = a$.
- Every element has an **inverse** a^{-1} : $a*a^{-1} = a^{-1}*a = e$.

These four properties, added one at a time, build up the algebraic structures below.

3. Semi-groups and Monoids

Semi-group. A **semi-group** is a set S together with an **associative** binary operation $*$. That is, $(a*b)*c = a*(b*c)$ for all $a, b, c \in S$.

Monoid. A **monoid** is a semi-group that also has an **identity element** e (with $a*e = e*a = a$ for all a).

Examples.

- $(\mathbb{N}, +)$ is a semi-group (associative); with 0 it is a monoid, but it is **not** a group (no inverses).
- $(\mathbb{Z}, \times)$ is a monoid (identity 1) but not a group (only ± 1 have inverses).

The ladder. Set + associative operation = **semi-group**; add identity = **monoid**; add inverses = **group**; add commutativity = **abelian group**.

4. Groups

Definition (Group). A **group** is an ordered pair $(G, *)$ where G is a set and $*$ is a binary operation on G such that:

1. **Closure:** $a * b \in G$ for all $a, b \in G$;
2. **Associativity:** $(a*b)*c = a*(b*c)$ for all $a, b, c \in G$;
3. **Identity:** there is an element $e \in G$ with $a * e = e * a = a$ for every $a \in G$;
4. **Inverse:** for every $a \in G$ there is an element $a^{-1} \in G$ with $a * a^{-1} = a^{-1} * a = e$.

Abelian (Commutative) Group. A group $(G, *)$ is **abelian** if the operation is also commutative: $a * b = b * a$ for all $a, b \in G$.

Order. The **order** of a group G , written $|G|$, is the number of elements in G . The **order of an element** a is the smallest positive integer n with $a^n = e$.

4.1 Examples and Non-examples

Groups.

- $(\mathbb{Z}, +)$: infinite abelian group (identity 0).
- $(\mathbb{Z}_n, +_n)$: finite abelian group of order n (identity 0).
- $(\mathbb{Q}^*, \times)$, $(\mathbb{R}^*, \times)$: non-zero numbers under multiplication.
- The fourth roots of unity $\{1, -1, i, -i\}$ under multiplication: an abelian group of order 4.
- The set of invertible 2×2 matrices under matrix multiplication (identity = I) — a **non-abelian** group.

Not groups. $(\mathbb{Z}, \times)$ is not a group (2 has no inverse). $(\mathbb{Z}, -)$ is not a group (not associative). $(\mathbb{N}, +)$ is not a group (no inverses).

4.2 Cayley Table of $(\mathbb{Z}_5, +_5)$

The operation table (Cayley table) of the group $(\mathbb{Z}_5, +_5)$:

$+_5$	0	1	2	3	4
0	0	1	2	3	4
1	1	2	3	4	0
2	2	3	4	0	1
3	3	4	0	1	2
4	4	0	1	2	3

Every entry is in $\{0,1,2,3,4\}$ (closure), the row/column of 0 repeats the header (identity 0), and every row contains every element exactly once (each element has an inverse).

5. Elementary Properties of Groups

Theorem (Basic properties). In a group $(G, *)$:

- The identity element is **unique**.
- Each element has a **unique** inverse.
- $(a^{-1})^{-1} = a$ and $(a*b)^{-1} = b^{-1} * a^{-1}$.
- **Cancellation:** $a*b = a*c \Rightarrow b = c$, and $b*a = c*a \Rightarrow b = c$.

Proof of uniqueness of identity. Suppose e and e' are both identities. Since e is an identity, $e * e' = e'$. Since e' is an identity, $e * e' = e$. Hence $e = e'$. So the identity is unique. $\square$

Proof of $(a*b)^{-1} = b^{-1}*a^{-1}$. Multiply: $(a*b)*(b^{-1}*a^{-1}) = a*(b*b^{-1})*a^{-1} = a*e*a^{-1} = a*a^{-1} = e$. So $b^{-1}*a^{-1}$ is the inverse of $a*b$. $\square$

Why this matters. Cancellation is what lets us "solve equations" inside a group, just like ordinary algebra.

6. Subgroups

Subgroup. A non-empty subset H of a group $(G, *)$ is a **subgroup** of G (written $H \leq G$) if H is itself a group under the same operation $*$.

Subgroup test (one-step). A non-empty subset $H \subseteq G$ is a subgroup if and only if for all $a, b \in H$, $a * b^{-1} \in H$. (Equivalently: H is closed under $*$ and under inverses.)

Examples.

- $n\mathbb{Z} = \{0, \pm n, \pm 2n, \dots\}$ is a subgroup of $(\mathbb{Z}, +)$.
- In $(\mathbb{Z}_6, +_6)$, the set $H = \{0, 3\}$ is a subgroup of order 2.
- Every group G has the **trivial subgroups** $\{e\}$ and G itself.

Every subgroup contains the identity and is closed under the operation — so $\{e\}$ is always the smallest subgroup.

7. Cyclic Groups

Cyclic Group. A group G is **cyclic** if there is an element $a \in G$ (a **generator**) such that every element of G is a power of a : $G = \langle a \rangle = \{a^n : n \in \mathbb{Z}\}$. The element a is said to **generate** G .

Theorem. Every cyclic group is **abelian**. (Because $a^m * a^n = a^{m+n} = a^n * a^m$.)

Examples.

- $(\mathbb{Z}_5, +_5)$ is cyclic, generated by 1: $\langle 1 \rangle = \{1, 2, 3, 4, 0\}$.
- $(\mathbb{Z}, +)$ is an infinite cyclic group generated by 1 (or by -1).

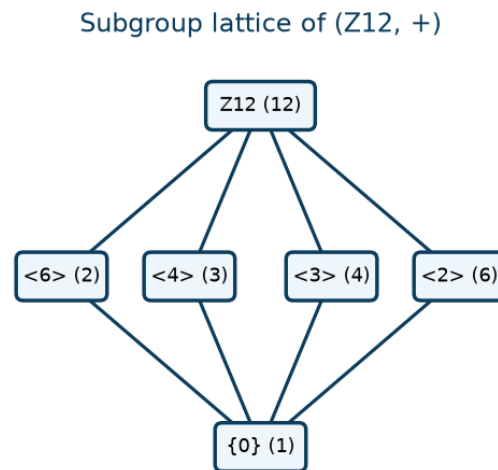


Fig. 7.1 — The subgroup lattice of $(\mathbb{Z}_{12}, +)$. Every subgroup order divides 12 (Lagrange's theorem).

8. Cosets

Coset. Let H be a subgroup of a group $(G, *)$ and let $a \in G$.

- The **left coset** of H determined by a is $aH = \{a * h : h \in H\}$.
- The **right coset** is $Ha = \{h * a : h \in H\}$.

Properties of cosets.

- $aH = H$ if and only if $a \in H$.
- Two left cosets are either **identical or disjoint**: $aH = bH$ or $aH \cap bH = \emptyset$.
- Every coset has the same size as H : $|aH| = |H|$.
- The distinct left cosets **partition** G .

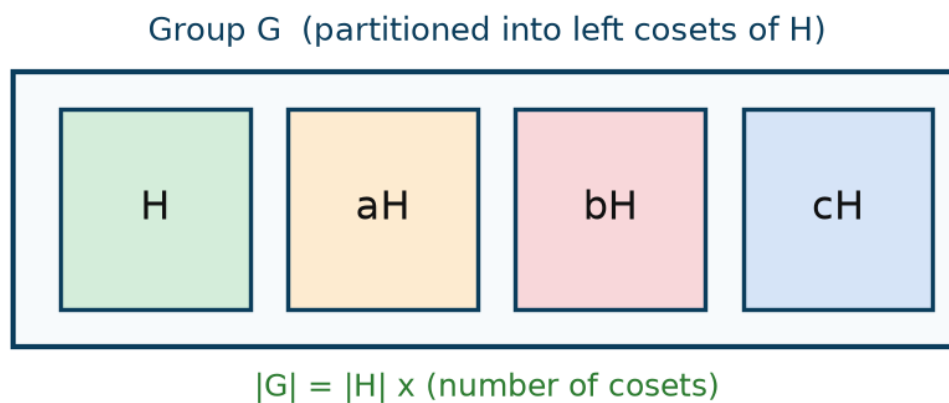


Fig. 8.1 — The group G is partitioned into disjoint left cosets of H , each of the same size as H .

Example. In $(\mathbb{Z}_6, +_6)$ with $H = \{0, 3\}$: the left cosets are $0+H = \{0, 3\}$, $1+H = \{1, 4\}$, $2+H = \{2, 5\}$ — three disjoint cosets, each of size 2.

9. Lagrange's Theorem

Lagrange's Theorem. If H is a subgroup of a finite group G , then the **order of H divides the order of G** : $|H| \mid |G|$. The number of distinct left cosets of H is the **index** $[G : H] = (|G|)/(|H|)$.

Why. The distinct left cosets partition G into blocks, each of size $|H|$ (Section 8). If there are k blocks, then $|G| = k \cdot |H|$. Hence $|H|$ divides $|G|$ and $k = |G|/|H|$. $\square$

Consequences.

- The order of every element divides $|G|$ (the element generates a cyclic subgroup).
- If $|G|$ is prime, the only subgroups are $\{e\}$ and G .

Example. If $|G| = 10$ and $|H| = 5$, then the number of distinct left cosets is $|G|/|H| = 10/5 = 2$.

10. Normal Subgroups and Quotient Groups

Normal Subgroup. A subgroup $H \leq G$ is **normal** (written $H \trianglelefteq G$) if its left and right cosets coincide for every $a \in G$: $aH = Ha$ for all $a \in G$. Equivalently, $aHa^{-1} \subseteq H$ for all $a \in G$.

Quotient (Factor) Group. If H is a normal subgroup of G , then the set of all cosets $G/H = \{aH : a \in G\}$ forms a group under the operation $(aH)*(bH) = (a*b)H$, called the **quotient group** (or factor group). Its order is $|G/H| = |G|/|H|$.

Key facts.

- Every subgroup of an **abelian** group is normal.
- H normal is exactly the condition needed for the cosets to form a group (otherwise $(aH)(bH)$ is not well-defined).

Example. In $(\mathbb{Z}_6, +_6)$ with $H = \{0, 3\}$ (which is normal, since the group is abelian), the quotient group $\mathbb{Z}_6 / H$ has $6/2 = 3$ elements: $\{H, 1+H, 2+H\}$ — isomorphic to $\mathbb{Z}_3$.

11. Homomorphism and Isomorphism

Homomorphism. Let $(G, *)$ and $(G', \circ)$ be groups. A function $\varphi : G \rightarrow G'$ is a **homomorphism** if it **preserves the operation**: $\varphi(a * b) = \varphi(a) \circ \varphi(b)$ for all $a, b \in G$.

Isomorphism. A homomorphism that is also a **bijection** (one-to-one and onto) is an **isomorphism**. If such a φ exists, G and G' are **isomorphic** ($G \cong G'$) — they are "the same group" with different labels.

Kernel and Image.

- **Kernel:** $\ker \varphi = \{a \in G : \varphi(a) = e'\}$ (the elements mapped to the identity).
- **Image:** $\text{Im } \varphi = \{\varphi(a) : a \in G\}$.

The kernel is always a normal subgroup of G , and $\text{Im } \varphi$ is a subgroup of G' .

Example. $\varphi : (\mathbb{Z}, +) \rightarrow (\mathbb{Z}_n, +_n)$, $\varphi(k) = k \bmod n$, is a homomorphism (since $(a+b) \bmod n = (a \bmod n) +_n (b \bmod n)$). Its kernel is $n\mathbb{Z}$.

Fundamental idea. Homomorphisms are "structure-preserving maps"; an isomorphism says two groups have exactly the same structure.

12. Coding of Binary Information and Error Detection

When binary data is transmitted or stored, bits can get corrupted. We fight this by **adding redundancy**: encoding short messages into longer **codewords** so that errors can be spotted.

Encoding Function. Let $B = \{0,1\}$ and B^n be the set of all n -bit strings. An **encoding function** is a function $e : B^k \rightarrow B^n$ ($n > k$) that maps a k -bit message to an n -bit **codeword**. The $n-k$ extra bits are **redundant (parity) bits**.

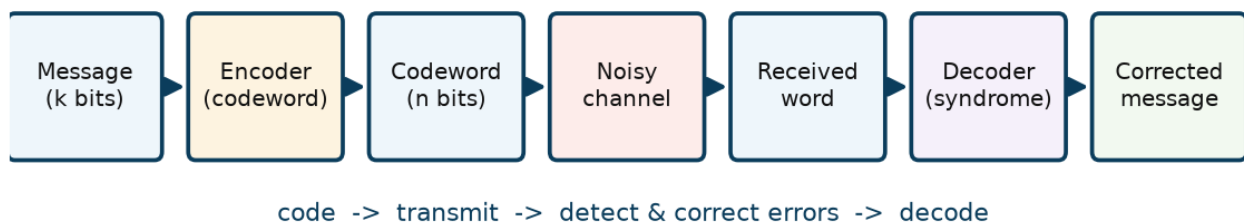


Fig. 12.1 — The coding chain: message $\rightarrow$ encode $\rightarrow$ transmit over a noisy channel $\rightarrow$ receive $\rightarrow$ decode & correct.

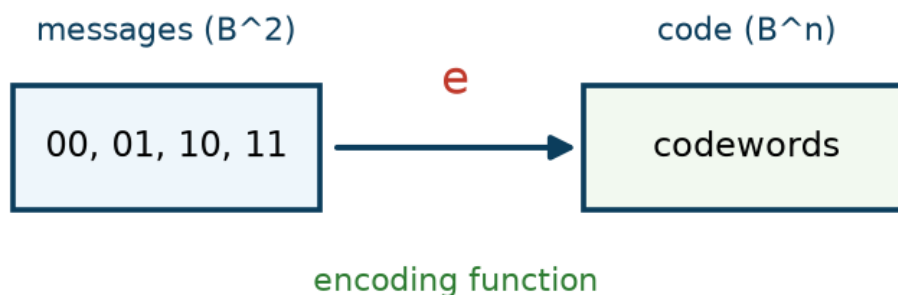


Fig. 12.2 — An encoding function maps the message space B^2 into a set of codewords in B^n .

Hamming Distance and Weight.

- The **Hamming distance** $d(x,y)$ between two binary words is the number of positions in which they differ.
- The **Hamming weight** $w(x)$ of a word is the number of 1s in it (its distance from the zero word).
- The **minimum distance** of a code is the smallest distance between any two distinct codewords.

Error capabilities. For a code with minimum distance d :

- it can **detect** up to $d - 1$ errors;
- it can **correct** up to $t = \lfloor (d-1)/2 \rfloor$ errors.

Parity-check example. Add one parity bit so the total number of 1s is even: $e(00)=000$, $e(01)=011$, $e(10)=101$, $e(11)=110$. A single flipped bit makes the parity odd, so one error is **detected** (but not located).

13. Group Codes

Group Code. A code is a **group code** if the set of all codewords forms a **group** under the bitwise XOR operation $\oplus$. Since XOR is its own inverse ($x \oplus x = 0$), a group code is exactly a set of codewords closed under $\oplus$, containing the zero word.

Key property of group codes. In a group code, the **minimum distance equals the minimum weight of a non-zero codeword**. This makes checking the error-correcting power very easy: just find the lightest non-zero codeword.

Example (the {000, 111} code). The codewords are 000 and 111. Check closure under $\oplus$: $000 \oplus 000 = 000$, $000 \oplus 111 = 111$, $111 \oplus 111 = 000$. Every result is again a codeword, so this set **is a group** under XOR — hence a group code. Its minimum weight is 3, so $d = 3$: it detects 2 errors and corrects 1.

Parity-Check Matrix and Generator Matrix.

- A **generator matrix** G (size $k \times n$) encodes a message $x \in B^k$ as the codeword $c = x G$.
- A **parity-check matrix** H (size $(n-k) \times n$) satisfies $G H^T = 0$; every codeword obeys $H c^T = 0$.

Example (a (2,5) linear code). Take $G = \begin{bmatrix} 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 \end{bmatrix}$. Then the codewords are 00 $\rightarrow$ 00000, 10 $\rightarrow$ 10110, 01 $\rightarrow$ 01011, 11 $\rightarrow$ 11101. Each non-zero codeword has weight 3, so $d = 3$.

14. Decoding and Error Correction

Decoding means: given a received word (which may contain errors), decide which codeword was most likely sent. For group codes this is done efficiently using the **syndrome**.

Syndrome. For a received word r and parity-check matrix H , the **syndrome** is $s = H r^T \pmod{2}$.

- If $s = 0$, then r is a valid codeword — no error detected.
- If $s \neq 0$, an error has occurred. For a **single** error, the syndrome equals the column of H corresponding to the error position.

Syndrome decoding algorithm.

1. Compute the syndrome $s = H r^T$.
2. If $s = 0$, accept r as the codeword.
3. Otherwise, find the column of H equal to s ; that column index gives the position of the (single) error.
4. Flip that bit and read off the original message from the corrected codeword.

Worked example. Using the (2,5) code above with parity-check matrix $H =$

$$\begin{bmatrix} 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \end{bmatrix}, \text{ suppose the received word is } r = 10010 \text{ (one error in the transmitted}$$

10110). The syndrome is $s = H r^T = (0,1,0)^T$, which matches column 4 of H — so bit 4 is wrong. Flipping it gives 10110, and the message is 10.

Maximum-likelihood decoding. The general (brute-force) method is to decode a received word to the **nearest codeword** (smallest Hamming distance). Syndrome decoding is the fast version of this for linear/group codes.

15. Question Bank — Unit V (Fully Solved)

Each solution states what is asked, the idea/tool used and why, the steps, and the final boxed answer.

Q1. Prove that Modulo 5 under addition $(\mathbb{Z}_5, +_5)$ forms a group. Further prove that the group is cyclic.

What is asked: verify the four group axioms for $(\mathbb{Z}_5, +_5)$, then exhibit a generator.

Group axioms.

- **Closure:** the Cayley table (Section 4.2) shows every $a +_5 b \in \{0,1,2,3,4\}$.
- **Associativity:** $(a+_5 b)+_5 c = a+_5 (b+_5 c)$ since ordinary addition is associative and taking mod 5 preserves it.
- **Identity:** 0 works, since $a +_5 0 = a$ for all a .
- **Inverse:** $0^{-1}=0, 1^{-1}=4, 2^{-1}=3, 3^{-1}=2, 4^{-1}=1$ (each pair sums to 5, i.e. 0 mod 5).

All four hold, so $(\mathbb{Z}_5, +_5)$ is a group.

Cyclic. Take the generator 1: its powers (multiples) are 1, 2, 3, 4, 0 mod 5, which is the whole group. So $\mathbb{Z}_5 = \langle 1 \rangle$ is cyclic (and abelian).

$(\mathbb{Z}_5, +_5)$ is a cyclic (hence abelian) group, generated by 1

Q2. What is a group code? Given the set of codewords $\{000, 111\}$, check whether this set forms a group under bitwise XOR.

Group code: a code whose codewords form a group under bitwise XOR $\oplus$ (closed under $\oplus$, with the zero word as identity).

Check closure for $\{000,111\}$: $000 \oplus 000 = 000$, $000 \oplus 111 = 111$, $111 \oplus 000 = 111$, $111 \oplus 111 = 000$. Every result is again a codeword, and 000 is the identity.

Yes — $\{000,111\}$ is a group code under XOR (minimum weight 3)

Q3. State Lagrange's Theorem.

Statement. If H is a subgroup of a finite group G , then $|H|$ divides $|G|$; the number of distinct left cosets is $[G:H] = |G|/|H|$.

Reason: the left cosets partition G into blocks each of size $|H|$.

$|H|$ divides $|G|$

Q4. If G is a group of order 10 and H is a subgroup of order 5, how many distinct left cosets does H have in G ?

By Lagrange's theorem the number of distinct left cosets is the index $[G:H] = (|G|)/(|H|) = 10/5 = 2$.

2 distinct left cosets

Q5. Construct a binary linear code using a generator matrix, and explain how syndrome decoding can correct errors.

Generator matrix (a (2,5) code): $G = \begin{bmatrix} 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 \end{bmatrix}$. A message x is encoded as $c = xG$,

giving codewords $00 \rightarrow 00000$, $10 \rightarrow 10110$, $01 \rightarrow 01011$, $11 \rightarrow 11101$ (each non-zero word has weight 3, so $d=3$ — corrects 1 error).

Parity-check matrix with $GH^T=0$: $H = \begin{bmatrix} 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \end{bmatrix}$.

Syndrome decoding. For a received word r compute $s = Hr^T$. If $s=0$, no error. Otherwise s equals the column of H at the error position; flip that bit to recover the codeword, then read the message.

$c=xG$; error located by the syndrome $s=Hr^T$

Q6. Show that $(G, \oplus_8)$ is an abelian group, where $G = \{0,1,2,3,4,5,6,7\}$.

Here $\oplus_8$ means addition modulo 8 (bitwise XOR on 3-bit strings is the same as addition mod 8 without carry... but as given, $\oplus_8$ is addition mod 8).

- **Closure:** $a +_8 b \in \{0, \dots, 7\}$ always.
- **Associativity:** inherited from integer addition.
- **Identity:** 0.
- **Inverse:** $a^{-1} = 8 - a$ (with $0^{-1}=0$), since $a + (8-a) = 8 \equiv 0$.
- **Commutative:** $a +_8 b = b +_8 a$.

All hold, so $(G, \oplus_8)$ is a group, and it is commutative — an abelian group.

$(\mathbb{Z}_8, +_8)$ is an abelian group

Q7. Show that the set of all positive rational numbers forms an abelian group under $a * b = ab/2$.

Let $G = \mathbb{Q}^+$ with $a*b = ab/2$.

- **Closure:** if $a, b > 0$ rational, then $ab/2$ is also a positive rational.
- **Associativity:** $(a*b)*c = ((ab/2)c)/2 = abc/4$ and $a*(b*c) = (a(bc/2))/2 = abc/4$ — equal.
- **Identity:** $e = 2$, since $a*2 = 2a/2 = a$ and $2*a = a$.
- **Inverse:** for a , take $a^{-1} = 4/a$; then $a * 4/a = (a \cdot 4/a)/2 = 2 = e$.
- **Commutative:** $a*b = ab/2 = ba/2 = b*a$.

So $(\mathbb{Q}^+, *)$ is an abelian group.

$(\mathbb{Q}^+, *)$ is an abelian group with identity 2 and $a^{-1} = 4/a$

Q8. Determine the (2,5) group code function e_H for the parity-check matrix H .

Idea. A (2,5) group code takes a 2-bit message $x = x_1x_2$ to a 5-bit codeword $e_H(x) = x_1x_2x_3x_4x_5$, where the parity bits x_3, x_4, x_5 are chosen so that $Hc^T = 0$. Using the H above, this gives $e_H(00) = 00000$, $e_H(01) = 01011$, $e_H(10) = 10110$, $e_H(11) = 11101$.

Check: each codeword satisfies $Hc^T = 0$; the non-zero codewords all have weight 3, so $d = 3$.

$e_H(00) = 00000$, $e_H(01) = 01011$, $e_H(10) = 10110$, $e_H(11) = 11101$

Q9. Show that the set of integers does not form a group under multiplication.

Check the axioms. $(\mathbb{Z}, \times)$ is closed (product of integers is an integer), associative, and has identity 1. But it fails the **inverse** axiom: for example, $2 \in \mathbb{Z}$ has no inverse in $\mathbb{Z}$, because we would need an integer b with $2b = 1$, and no such integer exists ($\nexists b \in \mathbb{Z}$).

$(\mathbb{Z}, \times)$ is not a group — elements like 2 have no inverse in $\mathbb{Z}$

Q10. Prove that the fourth roots of unity $\{1, -1, i, -i\}$ form an abelian multiplicative group.

Let $G = \{1, -1, i, -i\}$ under multiplication.

- **Closure:** e.g. $i \cdot i = -1$, $i \cdot (-i) = 1$, $(-i)(-i) = -1$ — all in G (in fact $G = \{i^n\}$).
- **Associativity:** complex multiplication is associative.
- **Identity:** 1.
- **Inverse:** $1^{-1} = 1$, $(-1)^{-1} = -1$, $i^{-1} = -i$, $(-i)^{-1} = i$.
- **Commutative:** complex multiplication is commutative.

So G is an abelian group of order 4 (cyclic, generated by i).

$\{1, -1, i, -i\}$ is a cyclic abelian group under multiplication

Q11. For the (2,4) encoding function $e(00) = 0000$, $e(10) = 0110$, $e(01) = 1011$, $e(11) = 1100$, determine how many errors can be detected and corrected.

Step 1 — minimum distance. Compute the weight of each non-zero codeword: $w(0110) = 2$, $w(1011) = 3$, $w(1100) = 2$. The minimum distance equals the minimum weight (a group code), so $d = 2$.

Step 2 — capabilities.

- Detect up to $d-1 = 1$ error.
- Correct up to $\lfloor (d-1)/2 \rfloor = \lfloor 1/2 \rfloor = 0$ errors.

Detects 1 error; corrects 0 errors ($d = 2$)

Q12. Check whether the set of all 2×2 matrices over $\mathbb{Q}$ under matrix multiplication forms a group.

Let $G = \{A : A \text{ is a } 2 \times 2 \text{ matrix over } Q\}$ under matrix multiplication.

- **Closure:** yes (product of two 2×2 matrices is 2×2).
- **Associativity:** yes (matrix multiplication is associative).
- **Identity:** yes (I).
- **Inverse: No.** A matrix with determinant 0 (e.g. the zero matrix) has no inverse. So not every element has an inverse.

Hence G is not a group. (The *invertible* 2×2 matrices do form a group, $GL(2, Q)$.)

No — singular matrices have no inverse, so the inverse axiom fails

Q13. Show that the (2,5) encoding function defined by $e(00)=00000$, $e(01)=01110$, $e(10)=10101$, $e(11)=11011$ is a group code.

Idea. A code is a group code iff its codewords are closed under bitwise XOR. Check every pair: $01110 \oplus 10101 = 11011$, $01110 \oplus 11011 = 10101$, $10101 \oplus 11011 = 01110$, $00000 \oplus c = c$ for all c , $c \oplus c = 00000$. Every result is again one of $\{00000, 01110, 10101, 11011\}$, and 00000 is the identity.

Minimum weight: the lightest non-zero codeword has weight 3, so $d = 3$ (detects 2 errors, corrects 1).

Yes — the codewords form a group under XOR, so it is a group code with $d=3$

16. Quick Revision Summary

Definitions to Remember

- **Binary operation** (closure); **semi-group** (associative); **monoid** (semi-group + identity).
- **Group**: closure + associativity + identity + inverses. **Abelian**: also commutative.
- **Order** of a group/element; **subgroup**; **cyclic group** (generator).
- **Coset** (aH , Ha); **index** $[G:H]$; **normal subgroup**; **quotient group** G/H .
- **Homomorphism** (operation-preserving map); **isomorphism** (bijective homomorphism); **kernel**.
- **Encoding function** $e:B^k \rightarrow B^n$; **codeword**; **Hamming distance/weight**; **group code**; **syndrome**.

Formulas & Results

Result	Statement
Lagrange's theorem	$ H $ divides $ G $
Number of cosets (index)	$[G:H] = G / H $
Order of quotient group	$ G/H = G / H $
Homomorphism	$\varphi(a*b) = \varphi(a) \circ \varphi(b)$
Errors detected (min distance d)	$d-1$
Errors corrected	$\lfloor (d-1)/2 \rfloor$
Group code min distance	= min weight of a non-zero codeword

Groups to Remember

Structure	Group?	Note
$(\mathbb{Z}, +)$	yes	infinite abelian, cyclic
$(\mathbb{Z}_n, +_n)$	yes	cyclic, order n
$(\mathbb{Z}, \times)$	no	no inverses
$(\mathbb{Q}^+, a*b=ab/2)$	yes	identity 2, $a^{-1}=4/a$
$\{1, -1, i, -i\}$ under $\times$	yes	cyclic order 4
2×2 matrices under $\times$	no	singular matrices fail

Key Points

- Every cyclic group is abelian; every subgroup of an abelian group is normal.
- In a group code, minimum distance = minimum weight of a non-zero codeword.
- A code with minimum distance d detects $d-1$ and corrects $\lfloor (d-1)/2 \rfloor$ errors.
- Syndrome $s = Hr^T$: zero $\Rightarrow$ no error; otherwise it points to the error position.

